Test Report issued under the responsibility of:

TEST REPORT
IEC 60825-1
Safety of laser products -
Part 1: Equipment classification and requirements

Report Number. ..... :	**SHES220300540501**
Date of issue..... :	**2022-05-12**
Total number of pages ..... :	14 pages
Name of Testing Laboratory preparing the Report ..... :	SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.
Applicant's name..... :	Richbeam (Suzhou) Technology Co., LTD
Address..... :	NO.788, Southeast Avenue, Changshu High Tech Industrial Development zone, Jiangsu, China
Test specification:	
Standard ..... :	IEC 60825-1:2014
Test procedure ..... :	CB Scheme
Non-standard test method ..... :	N/A
TRF template used ..... :	IECEE OD-2020-F1:2021, Ed.1.4
Test Report Form No. ..... :	IEC60825_1G
Test Report Form(s) Originator.... :	OVE
Master TRF..... :	Dated 2021-10-05

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Test item description..... :	2D TOF LIDAR
Trade Mark(s)..... :	RICHBEAM
Manufacturer..... :	Same as applicant
Model/Type reference..... :	LakiBeam 1, LakiBeam 1L
Ratings..... :	9-36 V d.c., Laser Class 1

Responsible Testing Laboratory (as applicable), testing procedure and testing location(s):

☒	**CB Testing Laboratory:**	SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.	
Testing location/ address.....:		588 West Jindu Road, Xinqiao, Songjiang, 201612 Shanghai, China.	
Tested by (name, function, signature).....:		Emilien Li Project Engineer	Emilien Li
Approved by (name, function, signature)...:		Nancy Zhang Reviewer	Nancy Z
☐	**Testing procedure: CTF Stage 1:**		
Testing location/ address.....:			
Tested by (name, function, signature).....:			
Approved by (name, function, signature)...:			
☐	**Testing procedure: CTF Stage 2:**		
Testing location/ address.....:			
Tested by (name + signature) .....:			
Witnessed by (name, function, signature)..:			
Approved by (name, function, signature)...:			
☐	**Testing procedure: CTF Stage 3:**		
☐	**Testing procedure: CTF Stage 4:**		
Testing location/ address.....:			
Tested by (name, function, signature).....:			
Witnessed by (name, function, signature)..:			
Approved by (name, function, signature)...:			
Supervised by (name, function, signature) :			

List of Attachments (including a total number of pages in each attachment):
-N/A

Summary of testing:

Tests performed	**Testing location:**
Full Test	SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd. 588 West Jindu Road, Xinqiao, Songjiang, 201612 Shanghai, China

Summary of compliance with National Differences (List of countries addressed):
-None

Use of uncertainty of measurement for decisions on conformity (decision rule) :

☒ No decision rule is specified by the IEC standard, when comparing the measurement result with the applicable limit according to the specification in that standard. The decisions on conformity are made without applying the measurement uncertainty ("simple acceptance" decision rule, previously known as "accuracy method").

☐ Other:… (to be specified, for example when required by the standard or client, or if national accreditation requirements apply)

Information on uncertainty of measurement:
The uncertainties of measurement are calculated by the laboratory based on application of criteria given by OD-5014 for test equipment and application of test methods, decision sheets and operational procedures of IECEE.
IEC Guide 115 provides guidance on the application of measurement uncertainty principles and applying the decision rule when reporting test results within IECEE scheme, noting that the reporting of the measurement uncertainty for measurements is not necessary unless required by the test standard or customer.
Calculations leading to the reported values are on file with the NCB and testing laboratory that conducted the testing.

Copy of marking plate:

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.

RICHBEAM

Product Name: 2D TOF LIDAR
Product Model : LakiBeam 1
S/N: RB001A27904182
MAC: 70:A6:C8:90:06:5B
MFG: 8/3/2022

RICHBEAM

Product Name: 2D TOF LIDAR
Product Model : LakiBeam 1L
S/N: RB001A27904183
MAC: 88:C9:B3:90:03:4E
MFG: 8/3/2022

Type 1
Supply Voltage: 9 V DC ... 36 V DC
Power Consumption: Typ. 2.5W
Aperture Angle: 270° (Horizontal)
Scanning Range: 20m (AT 10% Remission)
Response Time : Typ. 33 ms
Richbeam(Suzhou)Technology Co.,LTD
No.788,Southeast Avenue,Changshu high tech Industrial Development Zone,Jiangsu,China
www.richbeam.com
Class I Laser Product
IP65
MADE IN CHINA

Type 1
Supply Voltage: 9 V DC ... 36 V DC
Power Consumption: Typ. 2.5W
Aperture Angle: 270° (Horizontal)
Scanning Range: 15m (AT 10% Remission)
Response Time : Typ. 33 ms
Richbeam(Suzhou)Technology Co.,LTD
No.788,Southeast Avenue,Changshu high tech Industrial Development Zone,Jiangsu,China
www.richbeam.com
Class I Laser Product
IP65
MADE IN CHINA





Test item particulars .....:	--
Classification of installation and use.....:	--
Supply Connection .....:	--
.....:	
Possible test case verdicts:	
- test case does not apply to the test object..... :	N/A
- test object does meet the requirement..... :	P (Pass)
- test object does not meet the requirement..... :	F (Fail)
Testing ..... :	
Date of receipt of test item..... :	2022-03-24
Date (s) of performance of tests ..... :	2022-03-24 to 2022-05-12

General remarks:

"(See Enclosure #)" refers to additional information appended to the report.
"(See appended table)" refers to a table appended to the report.

Throughout this report a ☒ comma / ☐ point is used as the decimal separator.

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Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only.

Manufacturer's Declaration per sub-clause 4.2.5 of IECEE 02:

The application for obtaining a CB Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided.....:	☐ **Yes** ☒ **Not applicable**

When differences exist; they shall be identified in the General product information section.

Name and address of factory (ies)..... :	Richbeam (Suzhou) Technology Co., LTD NO.788, Southeast Avenue, Changshu High Tech Industrial Development zone, Jiangsu, China

General product information:

The product is a 2D TOF LIDAR using 940nm laser beam.
The model reference LakiBeam 1 is the reduced capacity version camparing to LakiBeam 1L due to software limitation.
The test in this report was conducted on LakiBeam 1L.

The product is classified to **class 1** laser product by the client because he considers all errors and statistical uncertainties that he knows from the production process and this is acceptable.

Following information should be on the manual:
a) adequate instructions for assembly, maintenance and safe use and description of the classification limitations, if appropriate.
b) additional warning for Class 1M and 2M
c) laser beam parameters for radiation above the AEL of Class 1 (Wavelength; Beam divergence; Maximum power or energy output)
d) safety instruction for embedded laser products and other incorporated laser products.
e) MPE and NOHD for Class 3B and 4 laser products; For collimated beam Class 1M and 2M lasers the extended NOHD (ENOHD).
f) information for the selection of eye protection.
g) reproduction of all required labels and warnings.
h) location of laser apertures
i) list of controls, adjustments of procedures for operation and maintenance - and warning statement.
j) information (compatibility requirements) about laser energy source if not incorporated.
k) additional warning for Class 1, 1M, 2, 2M, and 3R regarding skin or corneal burns.

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Clause	Requirement + Test	Result - Remark	Verdict

4	CLASSIFICATION PRINCIPLES		
4.3	Classification rules		---
4.3 a	Radiation of a single wavelength	940nm	P
4.3 b	Radiation of multiple wavelengths		N/A
	1) Laser product emits at two or more wavelengths shown as additive in Table 1		N/A
	2) Laser product emits at two or more wavelengths not shown as additive in Table 1		N/A
4.3 c	Radiation from extended sources (see 5.4.3)		N/A
4.3 d	Non-uniform, non-circular or multiple apparent source		N/A
4.3 e	Time bases		---
	1) 0,25 s		N/A
	2) 100 s	Class I	P
	3) 30000 s		N/A
4.3 f	Repetitively pulsed or modulated lasers		P
	1) Any single pulse	AEL single (J):$2*10^{-5}$ J	P
	2) Average power for pulse trains	AELT (J) : 0,039J	P
	3) Pulse duration $t \leq T_i$ ..... : Number of pulses N and C_5 ..... :		N/A
	3) Pulse duration $t > T_i$..... : Number of pulses N and C_5 ..... :	AEL s.p.train (J): $2*10^{-5}$ J	P
4.4	Laser products designed to function as conventional lamps.		N/A
	measured at 200 mm distance from closest point of human access (> 5 mrad).		N/A
	Un-weighted radiance L measured at 200 mm distance (comparison with L_T = 1 $MWm^{-2}sr^{-1}$/) under reasonably foreseeable single fault conditions.		N/A
	Evaluation of emission according to IEC 62471 series (optional): Standard applied (IEC 62471 series)……,…..... : Risk Group……..... : Labelling…… .. : Classification of product based on accessible laser radiation (if no laser radiation accessible: Class 1).		N/A

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Clause	Requirement + Test	Result - Remark	Verdict

5	**DETERMINATION OF THE ACCESSIBLE EMISSION LEVEL and PRODUCT CLASSIFICATION**		
5.1	Tests		---
	Compliance under reasonably foreseeable single fault conditions.	After evaluation on specifications, the EUT no higher emission under reasonably foreseeable single fault conditions	P
5.3	Determination of the class of the laser product ... : For Class 1C: vertical safety standard applied with requirements for Class 1C.		---
5.4	Measurement geometry		---
5.4.1	General		---
5.4.2	Default (simplified) evaluation		N/A
	Conditions applied ..... :		N/A
	Aperture diameter ..... :		N/A
	Reference point : ..... :		N/A
	Measurement distance ..... : (for each condition)		N/A
5.4.3	Evaluation condition for extended sources		P
	Conditions applied ..... :	Condition 3	P
	Most restrictive position ..... : (distance from reference point)	100 mm	P
	Angular subtense of the apparent source α and C_6: (for each condition)	C_6=1	P
5.4.3 a	Aperture diameters (for each condition). ..... :	Condition 3: 7mm	P
5.4.3 b	Angle of acceptance (for each condition)……...:		N/A

Measured accessible laser radiation and comparison with AEL:

Test result:

The measured peak wavelength is 940 nm.

After review, test result under condition 3 is stricter compared with condition 1.

Condition 3: $E_{single}=2,7*10^{-6}/30 = 9*10^{-8}$J

Class 1 limit: $2*10^{-5}$ J

Result doesn't exceed the limit of Class 1.

Therefore, the product has been classified to **Class 1** laser product.

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Clause	Requirement + Test	Result - Remark	Verdict

6	ENGINEERING SPECIFICATIONS		P
6.2	Protective housing		---
6.2.1	General		---
	Protective housing prevents access to energy levels in excess of the AEL for Class 1.		P
	Protective housing prevents access to energy levels equivalent to Class 4 and withstands exposures under reasonably foreseeable single fault conditions.		N/A
	Maintenance of Class 1, 1C, 1M, 2, 2M, or 3R (access to emissions of Class 3B or 4 is prevented).		P
	Maintenance of Class 3B product (access to emission of Class 4 is prevented).		N/A
6.2.2	Service		P
6.2.3	Removable laser system (laser system complies with requirements of Clauses 6 and 7).		N/A
6.3	Access panels and safety interlocks		---
6.3.1	Panel is intended to be removed during operation (or maintenance) and would give access to higher energy levels (see Table 13).		N/A
	Accessible emission (after removal of the panel) corresponds to product Class (designated by “X” in Table 13)		N/A
	Emission through the opening if interlocked panel of Class 1, 1C, 1M, 2, or 2M is removed (Emission < AEL of Class 1M or 2M).		N/A
	Emission through the opening if interlocked panel of Class 3R, 3B, or 4 is removed (Emission < AEL of Class 3R).		N/A
	Requirements regarding reasonably foreseeable single fault condition.		N/A
6.3.2	Override mechanism		N/A
	Behaviour of override in operation when the panel is replaced.		N/A
	Visible or audible warning for override mode.		N/A
6.4	Remote interlock connector		N/A
6.5	Manual reset		N/A
6.6	Key control		N/A
6.7	Laser radiation emission warning		---

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Clause	Requirement + Test	Result - Remark	Verdict
6.7.1	Laser product is a 3R (λ<400 nm; λ>700 nm), 1C, 3B or 4 laser systems.		N/A
6.7.2	Audible or visible warning.		N/A
	Warning is failsafe or redundant.		N/A
	Viewing of the visible warning does not require exposure to emissions > AEL for Class 1M and 2M.		N/A
6.7.3	Operational control and laser aperture are provided with a warning device when they are separated more than 2 m from warning device.		N/A
6.7.4	Visible indication of output aperture if laser emission may be distributed through more than one output.		N/A
6.7.5	Switch for handheld Class 3R device must be depressed for emission (in lieu of emission indicator).		N/A
6.8	Beam stop or attenuator		N/A
6.9	Controls		N/A
6.10	Viewing optics		N/A
	a) Human access to laser radiation in excess of Class 1M prevented when the shutter is opened or attenuation varied.		N/A
	b) Opening of the shutter or variation of the attenuation prevented when exposure to laser radiation in excess of Class 1M is possible.		N/A
6.11	Scanning safeguard		N/A
6.12	Safeguard for Class 1C products		N/A
	a) Human access to laser radiation in excess of AEL for Class 1 measured under Condition 3 is prevented.		N/A
	b) Human access to laser radiation in excess of AEL for Class 3B measured through 3,5 mm aperture at 5 mm distance from applicator is prevented.		N/A
6.13	Walk-in access		
	a) Means provided so that any person inside the housing can prevent activation of Class 3B or 4 laser hazards.		N/A
	b) A warning device provides adequate warning of emission to any person within the housing.		N/A
	c) Where “walk-in” access during operation is intended or reasonably foreseeable, emission of laser radiation that is equivalent to Class 3B or 4 while someone is present inside the enclosure of Class 1, Class 2 or Class 3R product is prevented by engineering means.		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
6.14	Environmental conditions		---
	- climatic conditions		N/A
	- vibration and shock		N/A
6.15	Protection against other hazards		---
6.15.1	Non-optical hazards (product safety standard)		N/A
	- electrical hazards;		N/A
	- excessive temperature;		N/A
	- spread of fire from the equipment;		N/A
	- sound and ultrasonics;		N/A
	- harmful substances;		N/A
	- explosion;		N/A
6.15.2	Collateral radiation		N/A
6.16	Power limiting circuit		N/A

7	**LABELLING**		
7.1	General		---
	Labels durable, permanently affixed		P
	Labels clearly visible		P
	Reading of labels is possible without exposure to laser radiation in excess of AEL for Class 1.		P
	Colour combination		N/A
	Labelling impractical due to the size or design of the product.		N/A
	Warning label – Hazard symbol (Figure 3)		N/A
7.2 - 7.7	Text on explanatory label or pictogram (laser class, warning text)		P
7.8	Aperture label		P
7.9	Radiation output and standards information		---
	Max output of laser radiation ..... :		N/A
	Pulse duration ..... :		N/A
	Emitted wavelength(s) ..... :		N/A
	Name and publication date of the standard ..... :		N/A
7.10	Labels for access panels		---
7.10.1 a) – f)	Labels for panels - warning wording used ..... :		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
7.10.2	Labels for safety interlocked panels - Warning wording used ..... :		N/A
7.11	Warning for invisible laser radiation ..... :		N/A
7.12	Warning for visible laser radiation ..... :		N/A
7.13	Warning for potential hazard to the skin or anterior parts of the eye - warning wording used..... :		N/A
8	**OTHER INFORMATIONAL REQUIREMENTS**		
8.1	Information for the user		---
	a) adequate instructions for assembly, maintenance and safe use and description of the classification limitations, if appropriate.		P
	b) additional warning for Class 1M and 2M		N/A
	c) laser beam parameters for radiation above the AEL of Class 1		---
	• Wavelength ..... :		N/A
	• Beam divergence ..... :		N/A
	• Pulse pattern ..... : (pulse duration, repetition rate, …)		N/A
	• Maximum power or energy output ..... :		N/A
	d) safety instruction for embedded laser products and other incorporated laser products.		N/A
	e) MPE and NOHD for Class 3B and 4 laser products; For collimated beam Class 1M and 2M lasers the extended NOHD (ENOHD).		N/A
	f) information for the selection of eye protection.		N/A
	g) reproduction of all required labels and warnings.		N/A
	h) location of laser apertures		N/A
	i) list of controls, adjustments of procedures for operation and maintenance - and warning statement.		N/A
	j) information (compatibility requirements) about laser energy source if not incorporated.		N/A
	k) additional warning for Class 1, 1M, 2, 2M, and 3R regarding skin or corneal burns.		N/A
	l) Information for Class 1C products (e.g. warning that repeated application may pose a risk).		N/A
8.2	Purchasing and service information		N/A

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Clause	Requirement + Test	Result - Remark	Verdict
	a) safety classification of each laser product stated in all descriptive material (e.g. brochures).		N/A
	b) adequate instructions for servicing available: • warnings and precautions regarding exposure of laser emission above Class 1 • maintenance schedule • list of controls and procedures that could increase accessible emissions • description of displaceable parts • protective procedures for service personnel • reproduction of labels and hazard warnings		N/A

9	**ADDITIONAL REQUIREMENTS FOR SPECIFIC LASER PRODUCTS**		
9.1	Applicable other parts of the standard series IEC 60825		---
	IEC 60825-2 (Safety of optical communication systems)		N/A
	IEC 60825-4 (Laser guards)		N/A
	IEC 60825-12 (Safety of free space optical communication systems used for transmission of information)		N/A
9.2	Medical laser products: Class 3B and Class 4 medical laser products comply with IEC 60601-2-22		N/A
9.3	Laser processing machines: Comply with IEC/ISO 11553 series.		N/A
9.4	Electric toys: Comply with IEC 62115		N/A
9.5	Consumer electronic products: Comply with IEC 60950 (IT-equipment) or IEC 60065 (AV equipment)		N/A

Attachment 1 Photos

Details of: overview

View:

[X] general

[ ] front

[ ] rear

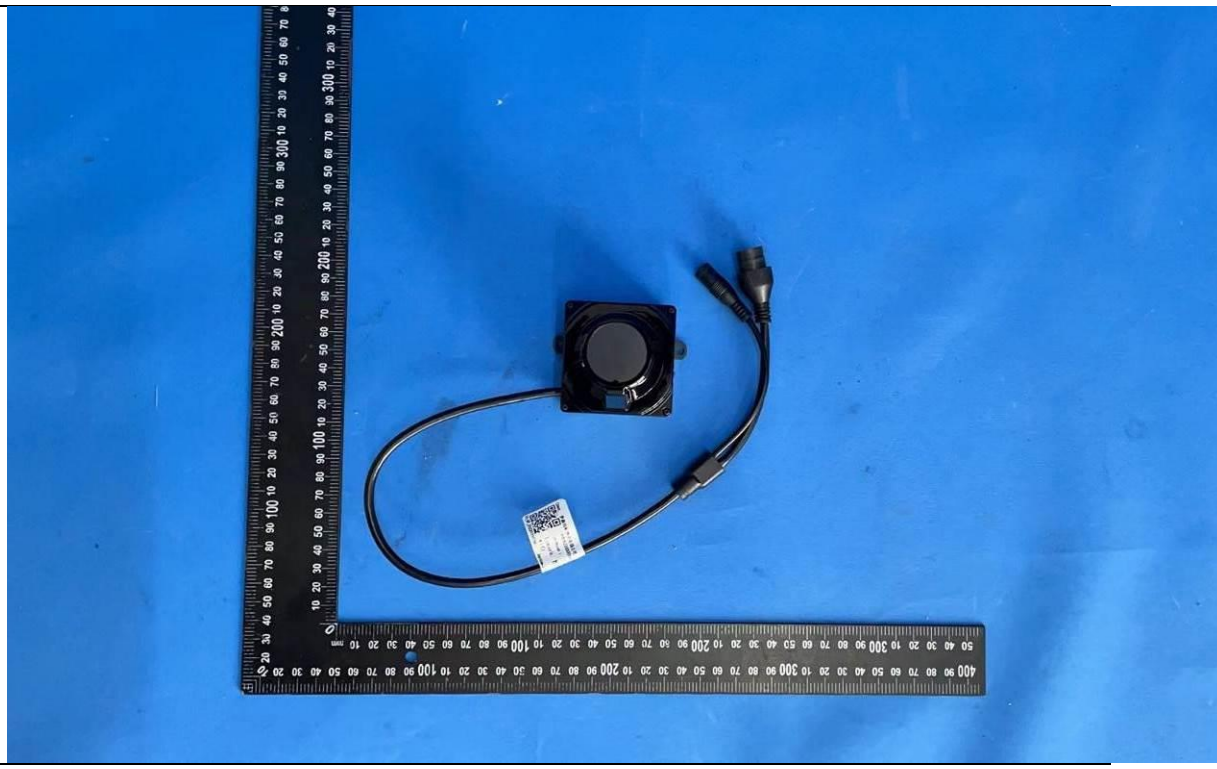
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Details of: overview

View:

[X] general

[ ] front

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---End of Report---